

D856

Environmental Management with Lab

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A teacher leads a stream-monitoring lab, using a new set of digital probes, on a creek near a closed quarry. Readings show 11 mg/L dissolved oxygen in March at 5 °C and 7 mg/L in August at 24 °C. The class collected samples at 9 a.m. both times. What best explains the drop?

- A. Summer algae always use up all the oxygen in a stream.
- B. Warmer water holds less dissolved oxygen.
- C. Warm water holds more oxygen, so the drop must be pollution.
- D. The closed quarry is releasing sediment that removes oxygen.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A power plant discharges heated cooling water into a river. What happens to dissolved oxygen just downstream?

- A. It decreases, because warmer water holds less dissolved oxygen.
- B. It stays the same, because no chemicals were added.
- C. It increases.
- D. It rises, because heat speeds up photosynthesis enough to outweigh solubility.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

Students testing rainwater in a school garden with 30 raised beds record a pH of 4.0; tap water from the same building reads 6.0. The garden uses drip irrigation. How many times higher is the rainwater's hydrogen-ion concentration?

- A. It is 100 times less acidic.
- B. 2 times
- C. 1.5 times
- D. 100 times

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A lake at pH 6.5 receives acid deposition that makes its hydrogen-ion concentration 10 times higher. What is its new pH?

- A.** 7.5
- B.** 5.5
- C.** 6.4
- D.** 0.65

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A county plans a new landfill on farmland next to a wetland. The project team, which has an award-winning design office, is drafting the environmental impact assessment. Which element is required for the EIA to be sound?

- A.** A plan to offset wetland loss elsewhere as the first step.
- B.** Baseline conditions measured before construction and a comparison of reasonable alternatives, including no action.
- C.** An assessment of impacts after the landfill opens.
- D.** A study of the preferred site only, since other sites cost more.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A home's electricity comes from a coal plant that is 35% efficient; transmission delivers 90% of that energy; an LED fixture converts 40% of its electricity to light. The home is on a street lined with oak trees. What percent of the coal's chemical energy becomes light?

- A.** About 12.6%
- B.** 31.5%
- C.** 165%
- D.** 55%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

An electric motor delivers 450 J of mechanical work for every 600 J of electrical energy it uses. What is its efficiency, and what happens to the rest?

- A.** 133%; the motor adds energy.
- B.** 25%
- C.** 75%; the other 150 J becomes mainly thermal energy.
- D.** 75%; the other 150 J is destroyed.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A mid-size city's treatment plant, which gives public tours each spring, passes wastewater through screens and settling tanks. The water still has a high biochemical oxygen demand (BOD) from dissolved organic matter. Which stage should come next?

- A.** Tertiary nutrient removal, skipping the microbial step.
- B.** Chlorination alone.
- C.** Secondary treatment: aerated tanks where microbes consume the dissolved organic matter.
- D.** More settling tanks.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

Volunteers from a local hiking club sample two stream sites with kick nets on a cool, cloudy morning. Site 1 has many mayfly, stonefly, and caddisfly larvae; Site 2 has mostly midge larvae and aquatic worms. Which conclusion is best supported?

- A.** Site 1 is completely free of pollution.
- B.** Site 2 is healthier, since worms signal rich conditions.
- C.** The sites are equal, since both support life.
- D.** Site 1 likely has better water quality, since mayflies, stoneflies, and caddisflies are pollution-sensitive.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A town wants to know whether a new fish-processing plant is affecting a river. The plant, which has 80 employees and a new logo, discharges at one outfall. The monitoring team has budget for two sampling sites. Where should they sample?

- A.** At the outfall pipe and at a clean lake in another watershed.
- B.** Two sites downstream of the outfall.
- C.** One site upstream of the outfall and one downstream past the mixing zone, using the same methods and timing.
- D.** Two sites upstream of the outfall.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A lab kit for a water-quality class includes a 1,000 mg/L nitrate stock solution in a bottle with a green cap. The instructor needs 500 mL of a 10 mg/L standard. How much stock should be diluted to a final volume of 500 mL?

- A.** 5 mL
- B.** 495 mL
- C.** 10 mL
- D.** 50,000 mL

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A 20 mL sample of a 250 mg/L sulfate solution is diluted to 100 mL. What is the new concentration?

- A.** 230 mg/L
- B.** 1,250 mg/L
- C.** 250 mg/L, since dilution does not change the amount of sulfate.
- D.** 50 mg/L

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A teacher leads a stream-monitoring lab, ~~using a new set of digital probes~~, on a creek ~~near a closed quarry~~. Readings show 11 mg/L dissolved oxygen in March at 5 °C and 7 mg/L in August at 24 °C. ~~The class collected samples at 9 a.m. both times.~~ What best explains the drop?

Target word: cause of the DO drop **Decides:** Water temperature rose from 5 °C to 24 °C.

A	Absolute language	Algae can affect DO, but 7 mg/L is not zero and “always” overreaches.
B	Correct	Gas solubility falls as temperature rises.
C	Inverted operation	Reverses the temperature-solubility relationship.
D	Camouflage	Nothing in the data links the quarry to the change.

2. Answer A Concept 1 • transfer

A power plant discharges heated cooling water into a river. What happens to dissolved oxygen just downstream?

Target word: effect of heat on DO **Decides:** Warmer water.

A	Correct	Thermal pollution lowers oxygen solubility and raises organisms’ oxygen demand.
B	Dropped element	Drops temperature as a factor; heat alone lowers DO.
C	Inverted operation	Warming lowers the amount of gas water can hold.
D	Confusion set	Mixes a biological effect with the physical rule; the solubility drop dominates.

3. Answer D Concept 2

Students testing rainwater in a school garden ~~with 30 raised beds~~ record a pH of 4.0; tap water from the same building reads 6.0. ~~The garden uses drip irrigation.~~ How many times higher is the rainwater’s hydrogen-ion concentration?

Target word: ratio of H⁺ **Decides:** Two pH units apart.

A	Inverted operation	A lower pH means more acidic, not less.
B	Wrong stem number	Uses the pH difference as the ratio.
C	Confusion set	Divides the pH values as if the scale were linear.
D	Correct	Each pH unit is 10×; $10 \times 10 = 100$.

4. Answer B Concept 2 • transfer

A lake at pH 6.5 receives acid deposition that makes its hydrogen-ion concentration 10 times higher. What is its new pH?

Target word: new pH from a concentration change **Decides:** 10× more H⁺ = 1 pH unit lower.

A	Inverted operation	Moves pH the wrong way; more H ⁺ lowers pH.
B	Correct	One factor of 10 lowers pH by 1.
C	Wrong stem number	Treats a 10-fold change as 0.1 pH unit.
D	Confusion set	Divides the pH by 10 as if pH were the concentration.

5. Answer B Concept 3

A county plans a new landfill on farmland next to a wetland. The project team, ~~which has an award-winning design office~~, is drafting the environmental impact assessment. Which element is required for the EIA to be sound?

Target word: required EIA element **Decides:** Baseline and alternatives.

A	Adjacent category	Offsetting is the last step of mitigation; avoidance comes first.
B	Correct	Impacts are judged against a baseline and other options.
C	Inverted operation	The EIA comes before the decision; after-opening checks are follow-up monitoring.
D	Dropped element	Drops the required comparison of alternatives.

6. Answer A Concept 4

A home's electricity comes from a coal plant that is 35% efficient; transmission delivers 90% of that energy; an LED fixture converts 40% of its electricity to light. ~~The home is on a street lined with oak trees.~~ What percent of the coal's chemical energy becomes light?

Target word: overall efficiency **Decides:** Multiply step efficiencies.

A	Correct	$0.35 \times 0.90 \times 0.40 = 0.126$.
B	Dropped element	Stops after the first two steps and drops the LED.
C	Inverted operation	Adds efficiencies; the total cannot exceed 100%.
D	Confusion set	Averages the three efficiencies.

7. Answer C Concept 4 • transfer

An electric motor delivers 450 J of mechanical work for every 600 J of electrical energy it uses. What is its efficiency, and what happens to the rest?

Target word: efficiency and energy accounting **Decides:** $450 \div 600$.

A	Inverted operation	Divides input by output.
B	Wrong stem number	That is the fraction lost, not the efficiency.
C	Correct	$450/600 = 0.75$; energy is conserved.
D	Confusion set	Energy is converted, not destroyed.

8. Answer C Concept 5

A mid-size city's treatment plant, ~~which gives public tours each spring~~, passes wastewater through screens and settling tanks. The water still has a high biochemical oxygen demand (BOD) from dissolved organic matter. Which stage should come next?

Target word: next treatment stage **Decides:** Dissolved organics remain after primary.

A	Confusion set	Tertiary targets nutrients like nitrogen and phosphorus, not the bulk BOD.
B	Dropped element	Disinfection kills pathogens but leaves the organic load.
C	Correct	Microbial treatment is what lowers BOD.
D	Adjacent category	Settling removes solids, not dissolved organics.

9. Answer D Concept 6

Volunteers ~~from a local hiking club~~ sample two stream sites with kick nets ~~on a cool, cloudy morning~~. Site 1 has many mayfly, stonefly, and caddisfly larvae; Site 2 has mostly midge larvae and aquatic worms. Which conclusion is best supported?

Target word: water quality from indicators **Decides:** EPT taxa at Site 1; tolerant taxa at Site 2.

A	Absolute language	Indicators suggest better quality, not zero pollution.
B	Inverted operation	Worms and midges dominate where tolerant species replace sensitive ones.
C	Dropped element	Drops which organisms are present.
D	Correct	Sensitive taxa need clean, well-oxygenated water.

10. Answer C Concept 7

A town wants to know whether a new fish-processing plant is affecting a river. The plant, ~~which has 80 employees and a new logo~~, discharges at one outfall. The monitoring team has budget for two sampling sites. Where should they sample?

Target word: sampling design **Decides:** One outfall and two sites.

A	Adjacent category	A different water body is not a fair reference.
B	Dropped element	Drops the reference site; nothing to compare against.
C	Correct	The upstream site is the reference for detecting the plant's effect.
D	Inverted operation	Neither site can show the discharge's effect.

11. Answer A Concept 8

A lab kit for a water-quality class includes a 1,000 mg/L nitrate stock solution ~~in a bottle with a green cap~~. The instructor needs 500 mL of a 10 mg/L standard. How much stock should be diluted to a final volume of 500 mL?

Target word: volume of stock **Decides:** $C_1V_1 = C_2V_2$.

A	Correct	$V_1 = (10 \times 500) \div 1,000 = 5 \text{ mL}$.
B	Confusion set	Subtracts volumes instead of using $C_1V_1 = C_2V_2$.
C	Wrong stem number	Uses the target concentration as a volume.
D	Inverted operation	Flips the ratio: $(1,000 \times 500) \div 10$.

12. Answer D Concept 8 · transfer

A 20 mL sample of a 250 mg/L sulfate solution is diluted to 100 mL. What is the new concentration?

Target word: concentration after dilution **Decides:** $C_2 = C_1V_1 \div V_2$.

A	Confusion set	Subtracts the volume from the concentration.
B	Inverted operation	Multiplies by 100/20 instead of 20/100.
C	Adjacent category	The amount stays the same, but it is spread through more volume.
D	Correct	$250 \times 20 \div 100 = 50 \text{ mg/L}$.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	4	
Adjacent category	The right idea in the wrong bucket.	4	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	11	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Dissolved oxygen and temperature	Cold water holds more dissolved oxygen than warm water, so DO falls in summer and below heated discharges even with no pollution.	Reading every seasonal DO drop as pollution, or thinking warm water holds more gas.	DO readings at two different water temperatures.
2	pH is logarithmic	Each 1-unit drop in pH is a 10-fold rise in hydrogen-ion concentration.	Treating pH differences as simple differences or ratios.	Two pH values and “how many times more acidic.”
3	Environmental impact assessment	A valid EIA sets baseline conditions before the project and compares reasonable alternatives, including no action, then plans mitigation (avoid first, then minimize, restore, offset).	Assessing only the preferred option or measuring impacts only after construction.	A project plan and a question about what the EIA must contain.
4	Energy conversion efficiency	Overall efficiency of a chain is the product of each step’s efficiency; the “lost” energy becomes mostly thermal energy, not destroyed.	Adding or averaging step efficiencies.	Several efficiencies in sequence.
5	Wastewater treatment stages	Primary treatment removes solids by screening and settling; secondary uses microbes to consume dissolved organics (lowers BOD); tertiary removes nutrients or other targets; disinfection kills pathogens.	Thinking disinfection or more settling removes dissolved organics.	High BOD remaining after settling.
6	Macroinvertebrate indicators	Mayfly, stonefly, and caddisfly larvae (EPT) are pollution-sensitive, so their abundance signals good water quality; dominance by tolerant midges and worms signals degradation.	Judging a stream by total life present, not which taxa.	A list of organisms found at stream sites.
7	Reference sites in monitoring design	To detect a source’s effect, sample upstream (reference) and downstream of it with the same methods and timing.	Sampling only downstream or using an unrelated water body as the reference.	One discharge point and limited sampling sites.
8	Dilution ($C_1V_1 = C_2V_2$)	When a solution is diluted, amount of solute stays the same: $C_1V_1 = C_2V_2$; mg/L $\approx$ ppm in water.	Flipping the ratio or subtracting volumes.	A stock concentration, a target concentration, and a final volume.

Course scope (public description): Environmental Management with Lab is an introductory course for undergraduate students seeking initial licensure or endorsement in secondary science education. This course provides a comprehensive overview of the principles and practices essential for understanding and managing the environment sustainably. This course focuses on sustainable use of energy, water, and other natural resources, as well as pollution of air, water, and land. Through hands-on labs and real-world examples, candidates will gain practical experience in environmental impact assessments and water and air quality monitoring. This course is

assessed via a Performance Assessment.